ST5202 — Applied Regression Analysis
Lecture Notes 6: Model Selection

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Outline

Background

Model selection criteria

Model selection procedures

1. Relevant and irrelevant predictors (1)

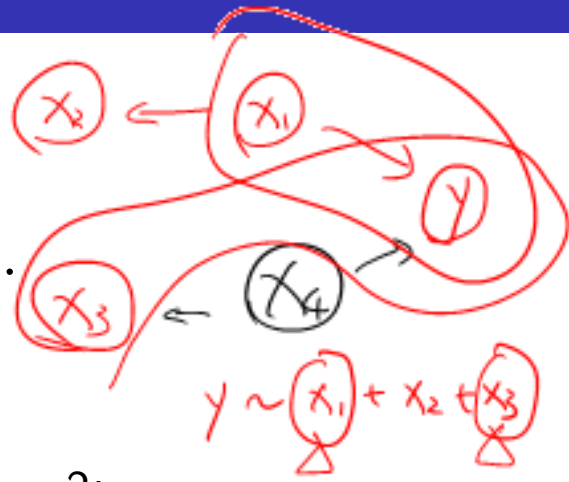
- In practical problems, there are usually a large number of possible predictors.
  - relevant predictors
  - irrelevant predictiors
- A *relevant predictor* is one that can explain a proportion of the variation of the response variable which cannot otherwise be explained by other predictors.
- A relevant predictor can be
  - a causal variable, i.e., the variation of the response is caused by the variation of the predictor
  - or a surrogate of certain causal variables. A surrogate variable represents a causal variable which is unobserved. The variation of the causal variable cause the variation of both the response and its surrogate.

2. Relevant and irrelevant predictors (2)

- An irrelevant predictor is neither a causal variable nor a surrogate of causal variable.

  It might have correlation with the response variable due to certain reasons. Such correlation is referred to as spurious correlation. That correlation might be because of
  - correlation with relevant predictors, e.g., both could be associated with other variables which are their cause.
  - data structure in the so-called small-n-large-p problem, where the number of possible predictors is larger than the number of observations.

3. An illustration of different kinds of predictors



- The data consists of observations

$$(y_i, x_{i1}, x_{i2}, x_{i3}), \quad i = 1, \ldots, 150.$$

- Generation model:

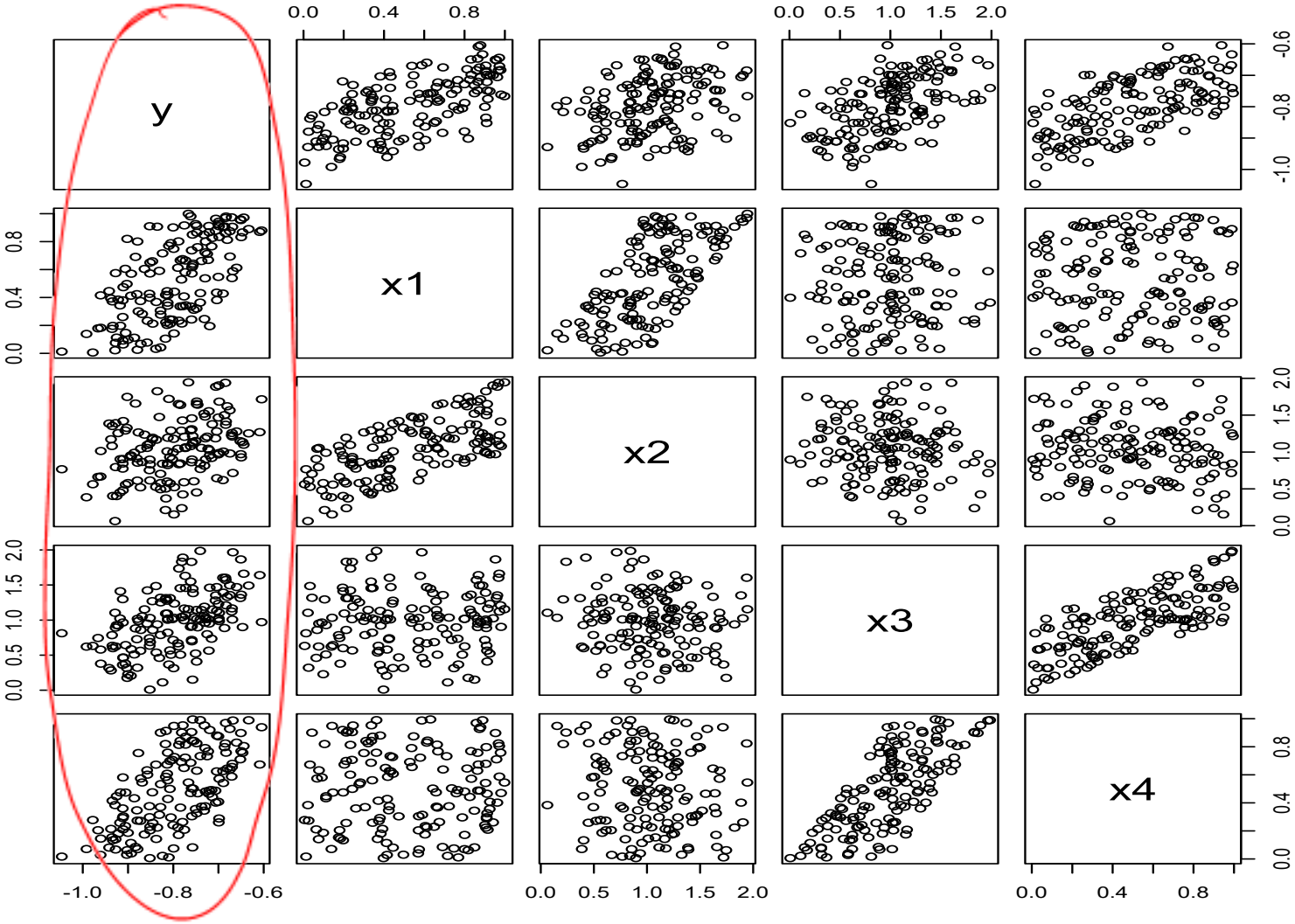
$$y \mid x_1, x_4 \sim N(-1 + 0.3x_1 + 0.2x_4, 0.2^2)$$

Furthermore, the predictors follow (all the uniform random variables are independent)

$$x_1 \sim \mathcal{U}[0,1], \quad x_2 = x_1 + \mathcal{U}[0,1], \quad x_3 = x_4 + \mathcal{U}[0,1], \quad x_4 \sim \mathcal{U}[0,1]$$

- Thus, x_1 is causal, x_2 is irrelevant, and x_3 is a surrogate of the causal variable x_4 (unobserved). There is a correlation between y and x_2 which is a spurious correlation.

4. Scatter plot matrix for y, x_1, x_2, x_3, x_4



5. Purpose of variable selection

- Model selection in most of case is equivalent to variable selection: identify the relevant predictors among a large number of candidate predictors.

  Specifically, we need to select predictors to fulfill one of the following goals:
  - Identify the relevant predictors, e.g., QTL mapping in genetic studies.
  - Build a model for prediction, e.g., disease diagnostics.
- The two goals have different emphases: the first emphasizes on the accuracy of **selection**; the second on the accuracy of **prediction**. The criteria is different.

$\{x_1, x_3\}$

$y = \hat{\beta}_1 x_1 + \hat{\beta}_3 x_3 + \hat{\beta}_0$

- Model selection approaches for the two goals are not completely different.

6. Under-fitting and over-fitting

Goal: the model under study consists of exactly the relevant predictors

$$y = \beta_0 + \beta_1 x_1 + \beta_3 x_3 + \epsilon.$$

- A model which does not contain all relevant predictors is called a under-fitting model

$$y = \beta_0 + \beta_1 x_1 + \epsilon.$$

- A model containing irrelevant predictors in addition to all the relevant predictors is called a over-fitting model.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon.$$

- Under-fitting and over-fitting have different effects on model selection.

7. Effect of under- and over-fitting on LSE (1)

$y = X\beta + \varepsilon$.

Let's start with a specific example to illustrate the effects

- Suppose that the design matrix is partitioned as $X = (X_1, X_2)$. Consider two models:

Overfitting $\Leftrightarrow$ Full Model: $$\boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon} = X_1\boldsymbol{\beta}_1 + X_2\boldsymbol{\beta}_2 + \boldsymbol{\epsilon}; \quad (1)$$

underfitting $\Leftrightarrow$ Reduced Model: $$\boldsymbol{y} = X_1\boldsymbol{\beta}_1 + \boldsymbol{\epsilon}. \quad (2)$$

Underfitting $\Leftrightarrow$ $\beta_2 \neq 0$

- Under the full model, we have (Lecture Notes 2)

$$\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{\beta}}_1^\top, \hat{\boldsymbol{\beta}}_2^\top)^\top = (X^\top X)^{-1} X^\top \boldsymbol{y}, \quad E\hat{\boldsymbol{\beta}} = \boldsymbol{\beta}, \quad \mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 (X^\top X)^{-1}.$$

By using the inverse of block matrix, it can be derived that

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}_1) = \sigma^2 [X_1^\top X_1 - X_1^\top X_2 (X_2^\top X_2)^{-1} X_2^\top X_1]^{-1}$$

- The LSE of $\boldsymbol{\beta}_1$ is always **unbiased** even it might be overfitting;

8. Effect of under- and over-fitting on LSE (2)

Underfitting $\Leftrightarrow \beta_2 \neq 0$
$\Leftrightarrow$ bias

- Under the reduced model, we have

$$\tilde{\boldsymbol{\beta}}_1 = (X_1^\top X_1)^{-1} X_1^\top \boldsymbol{y}, \quad \mathrm{Var}(\tilde{\boldsymbol{\beta}}_1) = \sigma^2 (X_1^\top X_1)^{-1}$$
$$E\tilde{\boldsymbol{\beta}}_1 = (X_1^\top X_1)^{-1} X_1^\top (X_1 \boldsymbol{\beta}_1 + X_2 \boldsymbol{\beta}_2) = \boldsymbol{\beta}_1 + (X_1^\top X_1)^{-1} X_1^\top X_2 \boldsymbol{\beta}_2.$$

- The LSE of $\boldsymbol{\beta}_1$ is unbiased only when $\boldsymbol{\beta}_2 = 0$, i.e. there is no underfitting.
- For the variance of the estimation of $\boldsymbol{\beta}_1$, we have

$$\underbrace{X_1^\top X_1}_{\text{Reduced}} > \underbrace{X_1^\top X_1 - X_1^\top X_2 (X_2^\top X_2)^{-1} X_2^\top X_1}_{\text{Full}}.$$

Smaller

(For two matrices, $A > B$ means that $A - B$ is positive definite.)
Hence, the reduced model always enjoys a **smaller variance**.

- Over-fitting increases variances; under-fitting increases biases.

9. Effect of under- and over-fitting on prediction

- Consider model M, where $\boldsymbol{y} = X_M\boldsymbol{\beta}_M + \boldsymbol{\epsilon}$. Let $\boldsymbol{\mu} = E[\boldsymbol{y}]$ denote the expectation, then the estimation of $\boldsymbol{y}$ is given by

$$\hat{\boldsymbol{y}}_M = X_M\hat{\boldsymbol{\beta}}_M = X_M(X_M^\top X_M)^{-1}X_M^\top \boldsymbol{y}$$
$$E\hat{\boldsymbol{y}}_M = X_M(X_M^\top X_M)^{-1}X_M^\top \boldsymbol{\mu},\ \mathrm{Var}(\hat{\boldsymbol{y}}_M) = \sigma^2 X_M(X_M^\top X_M)^{-1}X_M^\top,$$
$$\sum_{i=1}^n \mathrm{Var}(\hat{y}_{iM}) = \sigma^2\mathrm{tr}(X_M(X_M^\top X_M)^{-1}X_M^\top) = |M|\sigma^2,$$

where $|M|$ is the number of collumns of X, which is usually the number of predictors in model M plus 1.

$y_{n+1} \cdots y_{n+n}$

- The sum of prediction squared error (SPSE) on n unobserved y's is:

$$\begin{aligned}\text{SPSE} &= \sum_{i=1}^n E(y_{n+i} - \hat{y}_{iM})^2 = \sum_{i=1}^n E(y_{n+i} - \mu_i)^2 + \sum_{i=1}^n E(\mu_i - \hat{y}_{iM})^2 \\ &= n\sigma^2 + \sum_{i=1}^n E(\hat{y}_{iM} - \mu_{iM})^2 + \sum_{i=1}^n E(\mu_i - \mu_{iM})^2 \\ &= n\sigma^2 + |M|\sigma^2 + \sum_{i=1}^n (\mu_i - \mu_{iM})^2.\end{aligned}$$

$\sum_i E[\varepsilon_{n+i}^2]$ $\quad |M|$

$\mu_i = E[y_{n+i}]$
$\mu_{im} = E[\hat{y}_{im}]$

- (Bias-Variance Tradeoff) For prediction, again, over-fitting increases the variance, and under-fitting increases the bias

10. Principle of variable selection – Prediction focus

- The accuracy of an estimate $\hat{\beta}$ for a regression parameter β is measured by the mean squared error (MSE):

$$\begin{aligned} \text{MSE} &= E(\hat{\beta} - \beta)^2 = E(\hat{\beta} - E\hat{\beta} + E\hat{\beta} - \beta)^2 \\ \text{MSE}(\hat{\beta}) &= E(\hat{\beta} - E\hat{\beta})^2 + E(\beta - E\hat{\beta})^2 = \text{Var}(\hat{\beta}) + \text{bias}^2(\hat{\beta}). \end{aligned}$$

- To have a good prediction, we prefer M to strike a balance between the variance and bias, so that the SPSE or MSE can be minimized.
- In principle, SPSE or MSE should be used as criteria for variable selection, but they cannot be computed.
- Roughly speaking, a model selection criterion is *a surrogate of either SPSE or MSE*.

11. Selection criteria at a glance

Some typical criteria (not inclusive) are as follows.

- R^2 and R_a^2, suitable only for models with the same number of predictors.
- Mallow's C_p (complexity parameter), depends on an good estimate of σ^2.
- Akaike's information criterion (AIC), developed from the idea of minimizing the Kullback-Leibler distance between the current model and the "true" model.
- Bayesian information criterion (BIC), developed under a Bayesian framework.
- Extended Bayesian information criterion (EBIC), for models in small-n-large-p problems.
- Cross validation (CV), based on the estimated prediction error.

$n=100$.
10 relevant predictors
$p=20$

12. R^2 and R_a^2

- Recall that, for a linear model,

$$R^2 = \frac{\text{SSR}}{\text{SST}}, \quad R_a^2 = R^2 - \frac{p}{n-p-1}(1-R^2).$$

- R^2 and R_a^2 both are estimates of proportion of population variation of the response Y which is explained by the predictors in the regression model.
- For models with the same number of predictors, the larger the R^2 (R_a^2), the better the corresponding model.
- However, R^2 always increases as the number of predictors M increases and select the full model.

$y \sim x_1 + x_2$

$y \sim x_1 + x_3$

- Though R_a^2 is not strictly increasing as the number of predictors increases, it still increase when a predictor having even a very small contribution to the explained variation is added to the model.
- In general, both R^2 and R_a^2 are not proper criteria when models with different number of predictors are involved.

13. Mallow's C_p

recall: $$SPSE = \sum_{i=1}^{n} E(y_{n+i} - \hat{y}_{iM})^2 = n\sigma^2 + |M|\sigma^2 + \sum_{i=1}^{n}(\mu_i - \mu_{iM})^2.$$

intrinsic variance

- Since y_{n+i} are unavailable, an alternative is to use available data points, i.e., $\sum_{i=1}^{n}(y_i - \hat{y}_{iM})^2$. The calculation shows:

$$E\sum_{i=1}^{n}(y_i - \hat{y}_{iM})^2 = \text{SPSE} - 2|M|\sigma^2 \tag{3}$$

$\sum_{i=1}^{n}(y_i - \hat{y}_{iM})^2$ has a downward bias, due to data reuse

- Adjusting for the bias, we arrive at an unbiased estimate of SPSE:

$$\widehat{\text{SPSE}} = \sum_{i=1}^{n}(y_i - \hat{y}_{iM})^2 + 2|M|\hat{\sigma}^2.$$

- The Mallow's C_p is given by

$$C_p = \frac{1}{\hat{\sigma}^2}\sum_{i=1}^{n}(y_i - \hat{y}_{iM})^2 - n + 2|M|. \approx |M| + \frac{1}{\sigma^2}\sum(\mu_i - \mu_{iM})^2$$

where $\hat{\sigma}^2$ is an unbiased estimate of σ^2, and $-n$ is to remove the noise effect.

13. Mallow's C_p

recall: $$SPSE = \sum_{i=1}^{n} E(y_{n+i} - \hat{y}_{iM})^2 = n\sigma^2 + |M|\sigma^2 + \sum_{i=1}^{n} (\mu_i - \mu_{iM})^2.$$

- The Mallow's C_p is given by
$$C_p = \frac{1}{\hat{\sigma}^2} \sum_{i=1}^{n} (y_i - \hat{y}_{iM})^2 - n + 2|M|.$$
where $\hat{\sigma}^2$ is an unbiased estimate of σ^2, and $-n$ is to remove the noise effect.
- If model M is the correct model then $E[C_p] \approx |M|$.
- This criterion chooses the model with smallest value of C_p, which is closest to $|M|$.

14. Verification of formula (3)

$$
\begin{aligned}
& E\sum_{i=1}^{n}(y_i-\hat{y}_{iM})^2 \\
= \; & \underbrace{E\sum_{i=1}^{n}(y_i-\mu_i)^2+E\sum_{i=1}^{n}(\mu_i-\hat{y}_{iM})^2}_{\text{SPSE}} - 2E\sum_{i=1}^{n}(y_i-\mu_i)(\hat{y}_{iM}-\mu_i) \\
= \; & \text{SPSE} - 2E\sum_{i=1}^{n}(y_i-\mu_i)(\hat{y}_{iM}-\mu_i).
\end{aligned}
$$

$x_i (X^TX)^{-1}X^T y$

$\mathrm{cov}(y_i, \hat{y}_{iM})$

$$
\begin{aligned}
& E\sum_{i=1}^{n}(y_i-\mu_i)(\hat{y}_{iM}-\mu_i) \\
= \; & E\sum_{i=1}^{n}(y_i-\mu_i)\hat{y}_{iM} = \mathrm{tr}\big(E(\boldsymbol{y}-\boldsymbol{\mu})\hat{\boldsymbol{y}}_M^\top\big) = \mathrm{tr}\big(\mathrm{Cov}(\boldsymbol{y},\hat{\boldsymbol{y}}_M)\big) \\
= \; & \mathrm{tr}\big(\mathrm{Cov}(\boldsymbol{y}, X_M(X_M^\top X_M)^{-1}X_M^\top\boldsymbol{y})\big) = \sigma^2\mathrm{tr}(X_M(X_M^\top X_M)^{-1}X_M^\top) = |M|\sigma^2.
\end{aligned}
$$

15. Akaike's information criterion (AIC)

- True density: $f(y)$; Density under model M: $g_M(y)$.
  Kullback-Leibler distance measures the distance between densities:

$$I(f, g_M) = \int f(y) \log\left(\frac{f(y)}{g_M(y)}\right) dy = \int f(y) \log f(y) dy - \int f(y) \log g_M(y)$$

$$\approx \int f(y) \log f(y) dy - \frac{1}{n} \sum_{i=1}^{n} \log g_M(y_i).$$

constant

$\ll E[\log g_M(Y)] = \int (\log g_M(y)) \cdot f(y)\, dy$

- The Akaike's information criterion (AIC) essentially minimizes $I(f, g_M)$
- The first term is a constant, AIC approximates the second term, which has the form:

$$AIC = -2 \log L(\hat{\beta}_M, \hat{\sigma}_M^2) + 2j_M,$$

$j_M = p + 2$ (intercept, σ)

  where $\hat{\beta}_M, \hat{\sigma}_M^2$ are MLE under model M, L is the likelihood function, and j_M is the number of predictors in M.
- For multiple LRM with normality assumption, AIC is of the form:

$\hat{\beta}_M, \hat{\sigma}_M.$

$$AIC = n \log(\hat{\sigma}_M^2) + 2j_M + C,$$

  where $\hat{\sigma}_M^2 = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_{iM})^2$ and $C = n(\log(2\pi) + 1)$.

"bias" "var"

16. Bayesian infromation criterion (BIC)

- Let $\mathcal{M}$ be the set of all possible models, Pr a prior probability measure on $\mathcal{M}$. Let $Pr(M)$ be the prior on a model $M \in \mathcal{M}$ and $\pi(\boldsymbol{\beta}_M)$ the prior on parameters of model M.
- Given model M, the marginal density of data $\boldsymbol{y}$ is

$$m(\boldsymbol{y}|M) = \int L(\boldsymbol{y}; \boldsymbol{\beta}_M)\pi(\boldsymbol{\beta}_M)d\boldsymbol{\beta}_M.$$

- Therefore, the posterior probability of model M is

$$p(M|\boldsymbol{y}) = \frac{f(\boldsymbol{y}, M)}{f(\boldsymbol{y})} = \frac{m(\boldsymbol{y}|M)Pr(M)}{\sum_{M\in\mathcal{M}} Pr(M)m(\boldsymbol{y}|M)}.$$

- Let $Pr(M)$ be a constant (same prob. on every possible model), define BIC:

$$\Rightarrow \quad BIC = -2\log L(\hat{\boldsymbol{\beta}}_M, \hat{\sigma}^2_M) + j_M \log n.$$

AIC : $2 \cdot j_M$
BIC : $(\log n) \cdot j_M$

Then $BIC \approx -2\log p(M|\boldsymbol{y}) + C$, and minimizing BIC leads to a good model.

17. Extended Bayesian infromation criterion (EBIC)

- Partition $\mathcal{M}$ according to the number of predictors,

$$\mathcal{M} = \mathcal{M}_{,} \cup \mathcal{M}_{\infty} \cup \mathcal{M}_{\in} \cup \cdots$$

  where $\mathcal{M}_j$ is the set of all models consisting of j predictors. If Pr is the constant prior then

$$Pr(\mathcal{M}_j) \propto \binom{p}{j}$$

  where p is the total number of predictors.
- When p is large, $Pr(\mathcal{M}_j)$ with a big j is much larger than that with a small j. As a consequence, BIC tends to select models with more predictors.
- To address this problem, the EBIC puts a non-constant prior on the models:

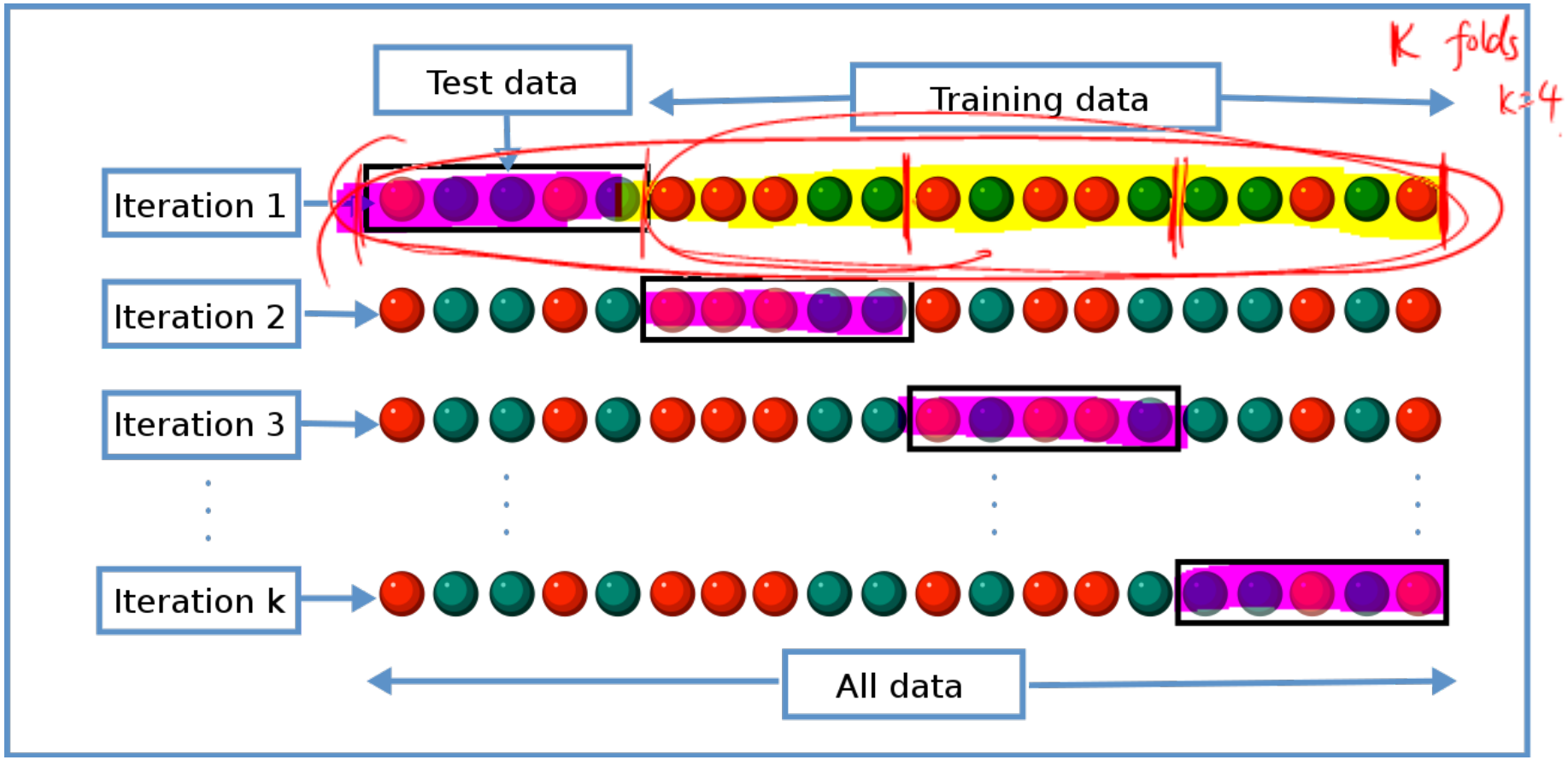
$$EBIC = -2\log L(\hat{\beta}_M, \hat{\sigma}_M^2) + j_M \log n + 2\gamma \log \binom{p}{j_M}, \ \gamma = \left(1 - \frac{\log n}{2\log p}\right)_+ .$$

- EBIC is developed in *Chen & Chen (2008) Biometrika, 95(3), 759-771*.

18. The idea of cross validation

- The above methods consider model selection based on $\boldsymbol{y}$ and X.
- When the sample size is sufficiently large, we can split it into two sets: training data and testing data. We fit the model on the training data and test the model on the testing data.
- However, it wastes information for inference. The statistical ideal is to estimate and assess the model by using all available information.
- The idea of cross-validation is to divide the data into different parts. Each part is alternatively reserved as testing data for the assessment of the model and the remaining parts are used as training data to estimate the model. The assessments are put together to form a score which approximates the prediction error.

19. Illustration of cross-validation



20. Leave-one-out CV and k-fold CV

- Leave-one-out (LOO) CV: the test data consists of only one observation and the training data consists of the remaining $n-1$ observations.
- k-fold CV: Divide the whole data into k parts. Each time, one part is used as the test data, and the remaining $k-1$ parts are used as training data.

- The k-fold CV has less computation demands, and LOO CV is more accurate.

21. The CV scores

For a model $y = \boldsymbol{x}^\top \boldsymbol{\beta} + \epsilon$, the CV scores are given below.

- LOO CV score:

$$CV = \frac{1}{n}\sum_{i=1}^{n}[y_i - \boldsymbol{x}_i^\top \hat{\boldsymbol{\beta}}^{(-i)}]^2,$$

  where $\hat{\boldsymbol{\beta}}^{(-i)}$ is the estimate of $\boldsymbol{\beta}$ obtained by leaving out the i-th data point $(y_i, \boldsymbol{x}_i)$.

- k-fold CV score:

$$CV_k = \frac{1}{n}\sum_{j=1}^{k}\|\boldsymbol{y}_j - X_j\hat{\boldsymbol{\beta}}^{(-j)}\|^2.$$

  where, $(\boldsymbol{y}_j, X_j)$ is the jth part of the data and is reserved as the testing data, and $\hat{\boldsymbol{\beta}}_{(-j)}$ is the estimate obtained by the remaining parts, the training data. The parts are randomly divided.

21. The CV scores (1)

- K-fold CV estimates that

  the prediction error when we have $\frac{K-1}{K}n$ data points

- However, we are interested in:

  the prediction error when we have n data points

- Adjust the systematic error

Adjusted CV = Raw CV + $[\frac{1}{n}\|\boldsymbol{y} - X\hat{\beta}\|^2$ - $\frac{1}{k}\sum_{j=1}^{k}\frac{1}{n}\|\boldsymbol{y} - X\hat{\beta}^{(-j)}\|^2]$.

22. An illustration

- Suppose we have a model $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$, and n observations $(y_1, x_{11}, x_{12}), \ldots, (y_i, x_{i1}, x_{i2}), \ldots, (y_n, x_{n1}, x_{n2})$.
- For each $i = 1, ..., n$, we use (y_i, x_{i1}, x_{i2}) as the test data, the remaining $n-1$ observations as the training data. Using the training data, we have the fitted regression function:

$$\hat{y} = \hat{\beta}_0^{(-i)} + \hat{\beta}_1^{(-i)} X_1 + \hat{\beta}_2^{(-i)} X_2.$$

  Using the model to predict y_i, we have the squared prediction error: $(y_i - \hat{\beta}_0^{(-i)} - \hat{\beta}_1^{(-i)} x_{i1} - \hat{\beta}_2^{(-i)} x_{i2})^2$.
- Taking the average of the squared prediction error over all i, we have the LOO CV score:

$$CV = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{\beta}_0^{(-i)} - \hat{\beta}_1^{(-i)} x_{i1} - \hat{\beta}_2^{(-i)} x_{i2})^2.$$

23. R functions for computing selection criteria

- The AIC and BIC can be computed by the functions `AIC` and `BIC` in the package `stats`. The syntax is as follows:

  ```
  AIC(object, ..., k = 2)
  BIC(object, ...)
  ```

  When k is set as $k = log(n)$, the function `AIC` computes BIC.

- The CV score can be computed by the function `cv.glm` in the package `boot`:

  ```
  cv = cv.glm(data, glmfit, cost, K)
  ```

  To use this function, the model must be fitted using the `glm` function which produces the same results as `lm` for linear models.

  The CV score is given by the `delta` component of `cv.glm`. It can be extracted as `cv$delta`, which is a vector of length two. The first component is the raw K-fold CV estimate of prediction error. The second component is the adjusted CV estimate (replacement of LOOCV).

24. Insurance Example (1)

- The data consists of nine variables — Y: survival time, X_1 : blood clotting score, X_2 : Prognostic index, X_3 : enzyme function test score, X_4 : liver function test score, X_5 : age in year, X_6 : indicator of gender (0=mail, 1=femail), X_7, X_8 indicator for alcohol use. The logarithm of Y is taken to be the response. The data is provided in the file `insurance.txt` on Canvas.
- Consider the full model
$$log.Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_6 X_6 + \beta_7 X_7 + \beta_8 X_8 + \epsilon,$$
and a reducedf model
$$log.Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_8 X_8 + \epsilon.$$
- R codes:

```
full.fit = glm(log.Y.~X1+X2+X3+X4+X5+X6+X7+X8,data=insur.dat)
CV.1.f = cv.glm(insur.dat,full.fit)$delta[2]
CV.k.f = cv.glm(insur.dat,full.fit,K=11)$delta[2]
aic.f = AIC(full.fit); bic.f = BIC(full.fit)

reduce.fit = glm(log.Y.~X1+X2+X3+X8,data=insur.dat)
CV.1.r = cv.glm(insur.dat,reduce.fit)$delta[2]
CV.k.r = cv.glm(insur.dat,reduce.fit,K=11)$delta[2]
aic.r = AIC(reduce.fit); bic.r = BIC(reduce.fit)
```

25. Insurance Example (2)

- The AIC, BIC and leave-out-one CV and 11-fold CV of the two models are computed as

```
Criterion  Full.Model Reduced.Model
     CV.1  0.05409740    0.05059544
     CV.k  0.05638025    0.04984915
      AIC -5.52562305   -8.10599197
      BIC 14.36421742    3.82791231
```

- If the computation of k-fold CV is repeated in R, each time the value will be different. It is because that each time the partition of the data set is different.

26. Remarks on the R functions

- In the functions `AIC` and `BIC` from the `stats` package, the number of parameters j_M is replaced by $j_M + 2$ which counts for the number of all parameters, including β_0, σ^2, and the un-reduced form of AIC and BIC is used, i.e., it includes the constant $n(\log(2\pi) + 1)$.
- Other packages (e.g. `VGAM`) might compute AIC and BIC differently in terms of the number of parameters and whether or not the constant $n(\log(2\pi) + 1)$ is included.
- The values of AIC and BIC for different models computed by different packages could differ. But the difference between values of different models are the same.

27. Model selection strategy

- A naive idea of model selection is to consider all possible models and then select among them by a selection criterion. Such method is called all subset selection.
- But the naive method is impractical. For p predictors, $X_1, ..., X_p$, there are

$$C_p^0 + C_p^1 + ... + C_p^p = 2^p$$

possible models. Even if p is not very large, the number 2^p is huge. For example, $2^{20} = 1,048,576$.
- The model selection strategy is as follow. (i) By using a certain mechanism, generate a small number of candidate models (relatively compared with 2^p); (ii) Using a certain selection criterion, select among the candidate models.

28. Procedure I: Removing redundant predictors

- This approach can be applied when p is not very large. There are two options of this approach:
  (i) Fit the full model, remove all predictors with a p-value bigger than a certain level α (say 0.05).
  (ii) Fit the full model, remove the predictor with the largest p-value which is bigger than α; then re-fit the model with the remaining predictors, repeat this procedure until none of the predictors has a p-value bigger than α.
- If the the covariates are not highly correlated, the two options produce the same selected model most of time. But when high correlation among the covariates exists, the second option is preferred.

29. Insurance Example revisited

- The full model considered is

$$log.Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_6 X_6 + \beta_7 X_7 + \beta_8 X_8 + \epsilon.$$

- The fitted results are

```
lm(formula = log.Y. ~ X1 + X2 + X3 + X4 + X5 + X6 + X7 + X8,
data = insur.dat)
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.050515   0.251756  16.089  < 2e-16 ***
X1           0.068512   0.025422   2.695  0.00986 **
X2           0.013452   0.001947   6.909 1.39e-08 ***
X3           0.014954   0.001809   8.264 1.43e-10 ***
X4           0.008016   0.046708   0.172  0.86450
X5          -0.003566   0.002752  -1.296  0.20163
X6           0.084208   0.060750   1.386  0.17253
X7           0.057864   0.067483   0.857  0.39574
X8           0.388383   0.088380   4.394 6.69e-05 ***
```

30. The selection results

- Set the level $\alpha = 0.05$.
- By the first option, the variables X_4 to X_7 are removed, since the p-values are smaller than 0.05.
- By the second option, the variables are removed in the order, X_4, X_7, X_5, X_6.
- Both options produces the same final model:

$$log.Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_8 X_8 + \epsilon$$

- The steps of the second option are illustrated in the following slides.

31. Model with X_4 removed

In the full model, x_4 has the larges p-value bigger than 0.05. It is removed first.

```
lm(formula = log.Y. ~ X1 + X2 + X3 + X5 + X6 + X7 + X8,
data = insur.dat)

             Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.036782   0.236169  17.093  < 2e-16 ***
X1           0.071434   0.018682   3.824 0.000394 ***
X2           0.013601   0.001722   7.896 4.21e-10 ***
X3           0.015150   0.001389  10.906 2.41e-14 ***
X5          -0.003709   0.002595  -1.429 0.159653
X6           0.087042   0.057842   1.505 0.139202
X7           0.058624   0.066624   0.880 0.383473
X8           0.388002   0.087415   4.439 5.62e-05 ***
```

32. Model with X_4, X_7 removed

In the previous model, x_7 has the largest p-value larger than 0.05, it is further removed.

```
lm(formula = log.Y. ~ X1 + X2 + X3 + X5 + X6 + X8,
data = insur.dat)

            Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.053974   0.234794  17.266  < 2e-16 ***
X1           0.071517   0.018637   3.837  0.00037 ***
X2           0.013755   0.001709   8.047 2.17e-10 ***
X3           0.015116   0.001385  10.912 1.78e-14 ***
X5          -0.003450   0.002572  -1.342  0.18620
X6           0.087317   0.057702   1.513  0.13691
X8           0.350904   0.076391   4.594 3.28e-05 ***
```

33. Model with X_4, X_7, X_5 removed

In the previous model, x_5 has the largest p-value larger than 0.05, it is further removed.

```
lm(formula = log.Y. ~ X1 + X2 + X3 + X6 + X8, data = insur.dat)

             Estimate Std. Error t value Pr(>|t|)
(Intercept) 3.867095   0.190572  20.292  < 2e-16 ***
X1          0.071241   0.018791   3.791 0.000419 ***
X2          0.013890   0.001721   8.073 1.71e-10 ***
X3          0.015115   0.001397  10.821 1.80e-14 ***
X6          0.086910   0.058180   1.494 0.141768
X8          0.362677   0.076515   4.740 1.94e-05 ***
```

34. Final model with X_4, X_7, X_5, X_6 removed

In the previous model, x_6 has the largest p-value larger than 0.05, it is further removed.

```
lm(formula = log.Y. ~ X1 + X2 + X3 + X8, data = insur.dat)

            Estimate Std. Error t value Pr(>|t|)
(Intercept) 3.852419   0.192695  19.992  < 2e-16 ***
X1          0.073323   0.018973   3.865 0.000327 ***
X2          0.014185   0.001731   8.196 9.58e-11 ***
X3          0.015453   0.001396  11.072 6.15e-15 ***
X8          0.352968   0.077191   4.573 3.29e-05 ***

Residual standard error: 0.2109 on 49 degrees of freedom
Multiple R-squared:  0.8299,    Adjusted R-squared:  0.816
F-statistic: 59.76 on 4 and 49 DF,  p-value: < 2.2e-16
```

None of the p-values is bigger than 0.05, the procedure stops.

35. Procedure II: Forward selection

$y \sim 1$; $y \sim x_2 + x_4$

- Forward selection is a sequential procedure. It starts with the NULL model M_0 with no predictors, then augments the model by including additional predictors one at a time, each time, the predictor having the largest contribution to reduce the residual sum of squares is added.
- The augmented model is compared with the previous model by a certain criterion. If the augmented model is favored by the criterion, the process is continued; otherwise, it is stopped and the previous model is taken as the final selected model.
- The criterion could be any one of the criteria discussed except R^2 and R_a^2. In what follows, the AIC is used.

36. Forward selection in detail

Suppose there are p predictors $X_1, \ldots, X_p$. Let AIC be the criterion. The procedure is as follows.

Step 1: Consider $Y = \beta_0 + \beta_j X_j + \epsilon,\ j = 1, \ldots, p$. Suppose X_{j_1} results in the smallest SSE. Designate the corresponding model as M_1. If $\text{AIC}(M_1) < \text{AIC}(M_0)$, continue; otherwise, stop.

step 2: Consider $Y = \beta_0 + \beta_{j_1} X_{j_1} + \beta_j X_j + \epsilon$ for all $j \neq j_1$. Suppose X_{j_2} results in the smallest SSE. Designate the corresponding model as M_2. If $\text{AIC}(M_2) < \text{AIC}(M_1)$, continue; otherwise, stop.

step k: Suppose $X_{j_1}, \ldots, X_{j_{k-1}}$ have been selected. Consider

$$Y = \beta_0 + \sum_{j=1}^{k-1} \beta_{j_k} X_{j_k} + \beta_j X_j + \epsilon,\ \text{all } j \neq j_1, \cdots, j_{k-1}.$$

Suppose X_{j_k} is the predictor such that the corresponding model has the smallest SSE. Designate the model as M_k. If $\text{AIC}(M_k) < \text{AIC}(M_{k-1})$, go to next step; otherwise, stop the procedure.

37. Insurance Example: forward selection (1)

Step 1:

```
Start:  AIC=-75.7
log.Y. ~ 1
```

$X_1, \cdots X_8$

	Df	Sum of Sq	RSS	AIC
+ X3	1	5.4762	7.3316	-103.827
+ X4	1	5.3990	7.4087	-103.262
+ X2	1	2.8285	9.9792	-87.178
+ X8	1	1.7798	11.0279	-81.782
+ X1	1	0.7763	12.0315	-77.079
+ X6	1	0.6897	12.1180	-76.692
<none>			12.8077	-75.703
+ X5	1	0.2691	12.5386	-74.849
+ X7	1	0.2052	12.6025	-74.575

38. Meaning of the results in each forward step

- `Sum of Sq`: The amount of regression sum of squares increased by adding the corresponding variable.
- `RSS`: The residual sum of square of the current model with the corresponding variable added.
- `AIC`: AIC of the current model computed by the formula $n\log(\hat{\sigma}^2_{mle}) + 2 * (m+1)$, where m is the number of variables in the current model.
- The `AIC` differ from that obtained by the AIC function in package `stats` which is computed using the formula $n\log(\hat{\sigma}^2_{mle}) + 2 * (m+2) + n(\log(2\pi) + 1)$.

39. Insurance Example: forward selection (2)

Step 2:

```
Step:  AIC=-103.83
log.Y. ~ X3

        Df Sum of Sq    RSS     AIC
+ X2     1   3.01908 4.3125 -130.48
+ X4     1   2.20187 5.1297 -121.11
+ X1     1   1.55061 5.7810 -114.66
+ X8     1   1.13756 6.1940 -110.93
<none>               7.3316 -103.83
+ X6     1   0.25854 7.0730 -103.77
+ X5     1   0.23877 7.0928 -103.61
+ X7     1   0.06498 7.2666 -102.31
```

$y \sim \underline{x_3}$

$x_1 \; x_2 \; x_4 \; x_5 \; \cdots \; x_8$?

$y \sim x_2 + x_3$

40. Insurance Example: forward selection (3)

Step 3:

```
Step:  AIC=-130.48
log.Y. ~ X3 + X2

       Df Sum of Sq    RSS     AIC
+ X8    1   1.46961 2.8429 -150.99
+ X1    1   1.20395 3.1085 -146.16
+ X4    1   0.69836 3.6141 -138.02
+ X7    1   0.22632 4.0862 -131.39
+ X5    1   0.16461 4.1479 -130.59
<none>              4.3125 -130.48
+ X6    1   0.08245 4.2300 -129.53
```

41. Insurance Example: forward selection (4)

Step 4:

```
Step:  AIC=-150.98
log.Y. ~ X3 + X2 + X8

       Df Sum of Sq    RSS     AIC
+ X1    1   0.66408 2.1788 -163.35
+ X4    1   0.46630 2.3766 -158.66
+ X6    1   0.13741 2.7055 -151.66
<none>              2.8429 -150.99
+ X5    1   0.07081 2.7721 -150.35
+ X7    1   0.02464 2.8182 -149.46
```

42. Insurance Example: forward selection (5)

Step 5:

```
Step:  AIC=-163.35
log.Y. ~ X3 + X2 + X8 + X1

       Df Sum of Sq    RSS     AIC
+ X6    1  0.096791 2.0820 -163.81
<none>              2.1788 -163.35
+ X5    1  0.075876 2.1029 -163.26
+ X4    1  0.041701 2.1371 -162.40
+ X7    1  0.022944 2.1559 -161.92
```

43. Insurance Example: forward selection (6)

Step 6:

```
Step:  AIC=-163.81
log.Y. ~ X3 + X2 + X8 + X1 + X6

       Df Sum of Sq    RSS     AIC
+ X5    1  0.076782 2.0052 -163.83
<none>              2.0820 -163.81
+ X7    1  0.022387 2.0596 -162.39
+ X4    1  0.016399 2.0656 -162.23
```

Step 7:

```
Step:  AIC=-163.83
log.Y. ~ X3 + X2 + X8 + X1 + X6 + X5

       Df Sum of Sq    RSS     AIC
<none>              2.0052 -163.83
+ X7    1  0.033193 1.9720 -162.74
+ X4    1  0.002284 2.0029 -161.90
```

44 Insurance Example: Final model by forward selection

```
lm(formula = log.Y. ~ X3 + X2 + X8 + X1 + X6 + X5,
data = insur.dat)
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.053974   0.234794  17.266  < 2e-16 ***
X3           0.015116   0.001385  10.912 1.78e-14 ***
X2           0.013755   0.001709   8.047 2.17e-10 ***
X8           0.350904   0.076391   4.594 3.28e-05 ***
X1           0.071517   0.018637   3.837  0.00037 ***
X6           0.087317   0.057702   1.513  0.13691
X5          -0.003450   0.002572  -1.342  0.18620
Residual standard error: 0.2066 on 47 degrees of freedom
Multiple R-squared: 0.8434, Adjusted R-squared: 0.8234
F-statistic: 42.2 on 6 and 47 DF, p-value: < 2.2e-16
```

45. Procedure III: Backward selection

- Backward selection is an inverted sequential method.
- It starts with the FULL model M_F with all the predictors, then reduces the model by removing predictors one at a time, each time, the predictor having the smallest contribution to reduce the residual sum of squares is removed.
- The reduced model is compared with the previous model by AIC. If the reduced model is favored by AIC, the process is continued; otherwise, it is stopped and the previous model is taken as the final selected model.

46. Backward selection in detail

Step 1: Consider $Y = \beta_0 + \sum_{l \neq j} \beta_l X_l + \epsilon,\ j = 1, \ldots, p$. Suppose the model with X_{j_1} removed has the smallest SSE. Designate the corresponding model as M_{-1}. If $\text{AIC}(M_{-1}) < \text{AIC}(M_F)$, continue; otherwise, stop.

step 2: Consider $Y = \beta_0 + \sum_{l \neq j, j_1} \beta_l X_l + \epsilon,\ j \neq j_1$. Suppose the model with X_{j_2} has the smallest SSE. Designate the corresponding model as M_{-2}. If $\text{AIC}(M_{-2}) < \text{AIC}(M_{-1})$, continue; otherwise, stop.

step k: Suppose $\mathcal{S}_{k-1} = \{X_{j_1}, \ldots, X_{j_{k-1}}\}$ have been removed. Consider

$$Y = \beta_0 + \sum_{l \notin \mathcal{S}_{k-1} \cup \{j\}} \beta_l X_l + \epsilon, \quad j \notin \mathcal{S}_{k-1}.$$

Suppose the model with X_{j_k} has the smallest SSE. Designate the model as M_{-k}. If $\text{AIC}(M_{-k}) < \text{AIC}(M_{-(k-1)})$, go to next step; otherwise, stop the procedure.

47. Insurance Example: backward selection (1)

Step 1:

```
Start:  AIC=-160.77
log.Y. ~ X1 + X2 + X3 + X4 + X5 + X6 + X7 + X8

        Df Sum of Sq    RSS     AIC
- X4    1   0.00129 1.9720 -162.74
- X7    1   0.03220 2.0029 -161.90
- X5    1   0.07354 2.0443 -160.79
<none>               1.9707 -160.77
- X6    1   0.08415 2.0549 -160.51
- X1    1   0.31809 2.2888 -154.69
- X8    1   0.84573 2.8165 -143.49
- X2    1   2.09045 4.0612 -123.72
- X3    1   2.99085 4.9616 -112.91
```

remove

$X_1 + X_2 + X_3 + X_5 + X_6 + X_7 + X_8$

48. Meaning of the results in each backward step

- `Sum of Sq`: The amount of regression sum of squares reduced by deleting the corresponding variable.
- `RSS`: The residual sum of square of the model eliminating all the deleted variables so far including the current corresponding variable.
- `AIC`: AIC of the above model.

49. Insurance Example: backward selection (2)

Step 2:

```
Step:  AIC=-162.74
log.Y. ~ X1 + X2 + X3 + X5 + X6 + X7 + X8

       Df Sum of Sq    RSS      AIC
- X7    1    0.0332 2.0052 -163.834
<none>               1.9720 -162.736
- X5    1    0.0876 2.0596 -162.389
- X6    1    0.0971 2.0691 -162.141
- X1    1    0.6267 2.5988 -149.833
- X8    1    0.8446 2.8166 -145.486
- X2    1    2.6731 4.6451 -118.471
- X3    1    5.0986 7.0706  -95.784
- X3    1   2.99085 4.9616 -112.91
```

50. Insurance Example: backward selection (3)

Step 3:

```
Step:  AIC=-163.83
log.Y. ~ X1 + X2 + X3 + X5 + X6 + X8

       Df Sum of Sq    RSS      AIC
<none>               2.0052 -163.834
- X5    1    0.0768 2.0820 -163.805
- X6    1    0.0977 2.1029 -163.265
- X1    1    0.6282 2.6335 -151.117
- X8    1    0.9002 2.9055 -145.809
- X2    1    2.7626 4.7678 -119.064
- X3    1    5.0801 7.0853  -97.672
```

The final selected model is the same as that selected by forward selection.

51. Procedure IV: Stepwise selection

- Stepwise selection is a mixture of forward and backward selection. It can be done in both directions: upwards and downwards.
- The upwards stepwise selection: In the forward selection procedure, once a new predictor is included into the model, the backward procedure is applied to the augmented model until no predictors can be removed, the procedure proceeds to the next forward step.
- The dowanwards stepwise selection: In the backward selection procedure, once a predictor is removed from the current model, a forward procedure is applied to the reduced model until no predictors can be added, the procedure proceeds to the next backward step.

52. Insurance Example: stepwise selection (1)

Step 1:

```
Start:  AIC=-75.7
log.Y. ~ 1

       Df Sum of Sq     RSS      AIC
+ X3    1    5.4762  7.3316 -103.827
+ X4    1    5.3990  7.4087 -103.262
+ X2    1    2.8285  9.9792  -87.178
+ X8    1    1.7798 11.0279  -81.782
+ X1    1    0.7763 12.0315  -77.079
+ X6    1    0.6897 12.1180  -76.692
<none>               12.8077  -75.703
+ X5    1    0.2691 12.5386  -74.849
+ X7    1    0.2052 12.6025  -74.575
```

53. Insurance Example: stepwise selection (2)

Step 2:

```
Step:  AIC=-103.83
log.Y. ~ X3

       Df Sum of Sq     RSS      AIC
+ X2    1    3.0191  4.3125 -130.483
+ X4    1    2.2019  5.1297 -121.113
+ X1    1    1.5506  5.7810 -114.658
+ X8    1    1.1376  6.1940 -110.932
<none>               7.3316 -103.827
+ X6    1    0.2585  7.0730 -103.765
+ X5    1    0.2388  7.0928 -103.615
+ X7    1    0.0650  7.2666 -102.308
- X3    1    5.4762 12.8077  -75.703
```

54. Insurance Example: stepwise selection (3)

Step 3:

```
Step:  AIC=-130.48
log.Y. ~ X3 + X2

        Df Sum of Sq    RSS      AIC
+ X8     1    1.4696 2.8429 -150.985
+ X1     1    1.2040 3.1085 -146.161
+ X4     1    0.6984 3.6141 -138.023
+ X7     1    0.2263 4.0862 -131.394
+ X5     1    0.1646 4.1479 -130.585
<none>               4.3125 -130.483
+ X6     1    0.0824 4.2300 -129.526
- X2     1    3.0191 7.3316 -103.827
- X3     1    5.6667 9.9792  -87.178
```

$Y \sim X_2 + X_3$

Adding a predicor in $X_1, X_4, \cdots X_8$

or removing a predictor in X_2, X_3

55. Insurance Example: stepwise selection (4)

Step 4:

```
Step:  AIC=-150.98
log.Y. ~ X3 + X2 + X8

       Df Sum of Sq    RSS      AIC
+ X1    1    0.6641 2.1788 -163.351
+ X4    1    0.4663 2.3766 -158.659
+ X6    1    0.1374 2.7055 -151.660
<none>               2.8429 -150.985
+ X5    1    0.0708 2.7721 -150.347
+ X7    1    0.0246 2.8182 -149.455
- X8    1    1.4696 4.3125 -130.483
- X2    1    3.3511 6.1940 -110.932
- X3    1    4.9456 7.7885  -98.562
```

56. Insurance Example: stepwise selection (5)

Step 5:

```
Step:  AIC=-163.35
log.Y. ~ X3 + X2 + X8 + X1

       Df Sum of Sq    RSS      AIC
+ X6    1    0.0968 2.0820 -163.805
<none>              2.1788 -163.351
+ X5    1    0.0759 2.1029 -163.265
+ X4    1    0.0417 2.1371 -162.395
+ X7    1    0.0229 2.1559 -161.923
- X1    1    0.6641 2.8429 -150.985
- X8    1    0.9297 3.1085 -146.161
- X2    1    2.9873 5.1661 -118.731
- X3    1    5.4513 7.6301  -97.671
```

57. Insurance Example: stepwise selection (6)

Step 6:

```
Step:  AIC=-163.81
log.Y. ~ X3 + X2 + X8 + X1 + X6

       Df Sum of Sq    RSS      AIC
+ X5    1    0.0768 2.0052 -163.834
<none>               2.0820 -163.805
- X6    1    0.0968 2.1788 -163.351
+ X7    1    0.0224 2.0596 -162.389
+ X4    1    0.0164 2.0656 -162.232
- X1    1    0.6235 2.7055 -151.660
- X8    1    0.9745 3.0565 -145.072
- X2    1    2.8268 4.9088 -119.490
- X3    1    5.0791 7.1611  -99.097
```

58. Insurance Example: stepwise selection (7)

Step 7:

```
Step:  AIC=-163.83
log.Y. ~ X3 + X2 + X8 + X1 + X6 + X5

       Df Sum of Sq    RSS      AIC
<none>              2.0052 -163.834
- X5    1    0.0768 2.0820 -163.805
- X6    1    0.0977 2.1029 -163.265
+ X7    1    0.0332 1.9720 -162.736
+ X4    1    0.0023 2.0029 -161.896
- X1    1    0.6282 2.6335 -151.117
- X8    1    0.9002 2.9055 -145.809
- X2    1    2.7626 4.7678 -119.064
- X3    1    5.0801 7.0853  -97.672
```

The stepwise selection also select the same model as that selected by forward selection.

59. R function for the sequential selections

- The R function `stepAIC` in the package `MASS` can be used to implement the forward, backward and stepwise selections.
- Usage:

```
object = lm(y~ , ...)
stepAIC(object, scope,  direction =
c("both", "backward", "forward"), ...)
```

- `object` is the fitted starting model, for forward selection, the starting model is the NULL model, for backward selection, the starting model is the Full model. For stepwise selection, it could be either one of them, but the starting model determines the direction of the stepwise selection.
- `scope` specifies the lower and upper ends of the models.
- `direction` specifies the procedure.

60. Pull strength Example

- The pull strength of a wire bond is an important characteristic. The data set `pullstrength.csv` provided in Canvas gives information on pull strength (y), die height (x_1), post height (x_2), loop height (x_3), wire length (x_4), bond width on the die (x_5) and bond width on the post (x_6).
- The following R codes are used for the model selection:

```
pull.dat=read.csv(file="D://Rsessions/pullstrength.csv",
header=TRUE)
fit.1 = lm(y~1, data=pull.dat)
fit.2 = lm(y~x1+x2+x3+x4+x5+x6, data=pull.dat)
forward=stepAIC(fit.1,scope=list(upper=~x1+x2+x3+x4+x5+x6,
lower=~1),direction="forward")
backward=stepAIC(fit.2,scope=list(upper=~x1+x2+x3+x4+x5+x6,
lower=~1),direction="backward")
stepwiseF=stepAIC(fit.1,scope=list(upper=~x1+x2+x3+x4+x5+x6,
lower=~1),direction="both")
stepwiseB=stepAIC(fit.2,scope=list(upper=~x1+x2+x3+x4+x5+x6,
lower=~1),direction="both")
```

61. Pull strength Example (cont.)

- All the four procedures produce the same final model.
- The result of the final model can be extracted using the function `summary()`, say `summary(forward)` gives the following result:

```
lm(formula = y ~ x3 + x4, data = pull.dat)

Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.65631    6.39482   0.728  0.47705
x3           0.51133    0.13075   3.911  0.00125 **
x4          -0.12418    0.04466  -2.781  0.01336 *

Residual standard error: 0.8554 on 16 d.f.
Multiple R-squared:0.6476, Adjusted R-squared:0.6035
F-statistic: 14.7 on 2 and 16 DF, p-value: 0.0002379
```

62. Penalized likelihood approach

AIC $-2\log$ Likelihood $+ J_m \cdot 2$

- For a LRM $\boldsymbol{y} = X\boldsymbol{\beta} + \epsilon$, the penalized likelihood approach select variables by minimizing

$$\frac{1}{n}\|\boldsymbol{y} - X\boldsymbol{\beta}\|_2^2 + p_\lambda(\boldsymbol{\beta}),$$

$\frac{1}{n}\|y - \hat{f}(x)\| + P_\lambda(\hat{f})$

$P_\lambda(\beta) = \lambda \cdot \|\beta\|_0$

Lasso: $\|\beta\|_1$

Ridge: $\|\beta\|_2^2 = \sum \beta_i^2$

where p_λ is called the penalty function, λ is the penalty parameter whose value is to be chosen.

- The procedure for the penalized likelihood approach to select variables is essentially as follows: first specify a sequence of λ values, then, at each value, carry out the penalized minimization, which yields a model with certain selected variables. Finally, a selection criterion is used to select the models.
- The penalty parameter λ can be chosen by a certain criterion depending on the purpose of the study. If it is to obtain a model for prediction, CV can be used. If it is to identify important variables, EBIC can be used.

63. Common penalty functions

- Ridge penalty: $p_\lambda(\boldsymbol{\beta}) = \lambda \sum_{j=1}^{p} \beta_j^2$. There is a shrinkage effect for every coefficient, but not variable selection.
- LASSO penalty: $p_\lambda(\boldsymbol{\beta}) = \lambda \sum_{j=1}^{p} |\beta_j|$. LASSO combines variable selection and shrinkage. It suffers from bias and instability.
- Adaptive Lasso penalty: $\lambda \sum_{j=1}^{p} w_j |\beta_j|$, where w_j is taken as $|\hat{\beta}_j|^{-1}$. $\hat{\beta}_j$ is the OLS estimate if $p \ll n$ and the OLS estimate in the marginal LRM when p and n are close.
- SCAD: $p_\lambda(|\beta|) = |\beta|$ for $|\beta|$ near 0, and equals a constant C for large $|\beta|$, with a smooth connecting function. It corrects the strong penalty on true non-zero coefficients. Large computation cost.
- MCP: Similar with SCAD, with a more direct expression. Large computation cost.

$(X^TX)^{-1}X^Ty$

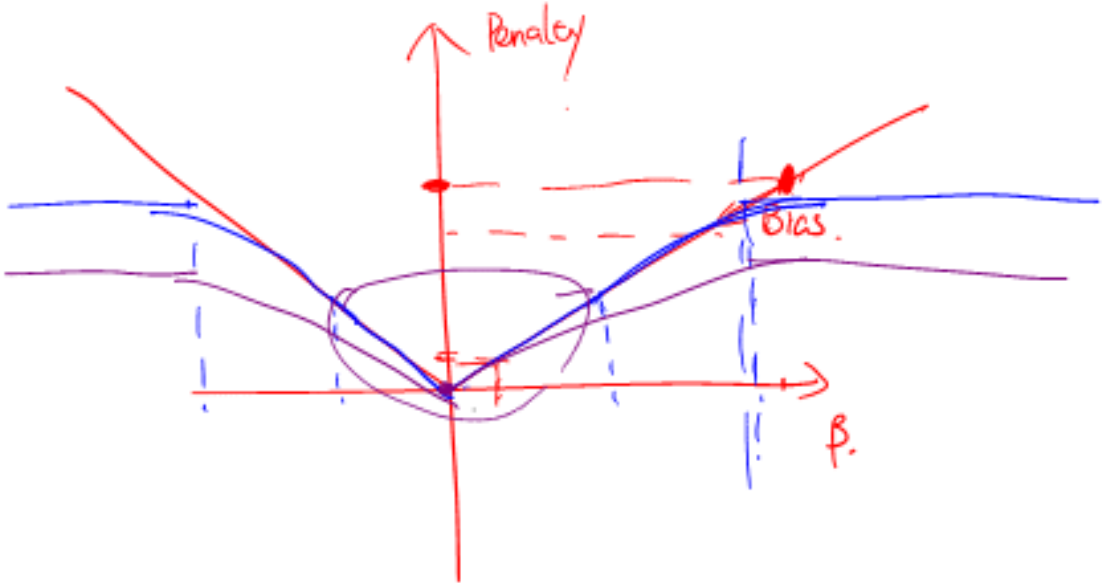
$(X^TX+\lambda I)^{-1}X^Ty$

$\min \frac{1}{n}(y-X\beta)^T(y-X\beta) + \lambda\beta^T\beta$

Lasso : $(\hat{\beta}_{LSE})_j$ small $\xrightarrow{\text{Lasso}}$ 0

$(\hat{\beta}_{LSE})_j$ large $\xrightarrow{\text{lasso}}$ $(\hat{\beta}_{LSE})$ · Shrinkage

Penalty $\|\beta\|_1$



Adaptive lasso:

$$\lambda \sum_{j=1}^{p} |\beta_j| \Rightarrow \lambda \sum_{j=1}^{p} w_j |\beta_j|$$

$$w_j = 1/|\hat{\beta}_j|$$

SCAD : γ

~~MCP~~

MCP : γ

64. Rational of penalized likelihood approach

- Take LASSO (Least absolute shrinkage and selection operator) as the example. The LASSO estimates $\boldsymbol{\beta}$ by minimizing

$$\frac{1}{n}\|\boldsymbol{y} - X\boldsymbol{\beta}\|_2^2 + \lambda \sum_{j=1}^{p} |\beta_j|.$$

- If $\lambda = 0$, the LASSO estimator is the same as the LSE. overfitting
- If $\lambda = \infty$, all the components of $\boldsymbol{\beta}$ are estimated as 0
- For a certain nonzero λ, some of the components will be estimated as nonzero, and the others 0. The nonzero ones are shrunk version of LSE.
- The variables with nonzero estimated coefficients are the selected variables.

Penalized regression: a comparison

Method	Sparse	Bias[1]	Convex	Prediction	Selection
Ridge	No	Yes	Yes	Yes	No
LASSO	Yes	Yes	Yes	Yes	Yes
AdaLASSO	Yes	Less than LASSO	Yes	Yes	Yes
SCAD	Yes	Small	No	Yes	Yes
MCP	Yes	Small	No	Yes	Yes

[1] Bias here refers to bias for large coefficients.

65. R packages for penalized likelihood approaches

- There are many R packages for the implementation of penalized likelihood approaches, which can be searched on the internet.
- We are going to use the package `glmnet` for the implementation of LASSO.
- The `glmnet` package requires the following packages: `iterators`, `foreach`, `Matrix` and `shape`.
- The function `cv.glmnet` in the `glmnet` package can be used as follows:

```
fit = cv.glmnet(x, y)
coef(fit, s = fit$lambda.min)
```

where `y` is the response vector, `x` is the design matrix without the column **1**, `fit$lambda.min` is the λ value producing the smallest CV score.

66. Insurance example (cont.)

The R code for the implementation of LASSO:

```
library(iterators)
library(foreach)
library(Matrix)
library(shape)
library(glmnet)
x = as.matrix(insur.dat[,1:8])
y = as.vector(insur.dat[,10])
fit1 = cv.glmnet(x, y)
coef(fit1, s = fit1$lambda.min)
```

67. Insurance example (cont.)

- The selected variables and their coefficients:

```
(Intercept)  4.086623625
X1           0.064558030
X2           0.013208367
X3           0.014653338
X4           0.014324205
X5          -0.003161543
X6           0.077361533
X7           0.044432679
X8           0.374970854
```

- Note: the estimated coefficients are not the LSE. They are in general shrunk estimates compared with LSE (except for X4, all the other coefficients have smaller absolute values that those of LSE).

68. A simulated example

- R code for the simulation:

```
set.seed(1010)
n = 1000
p = 100
nzc = trunc(p/10)
x = matrix(rnorm(n * p), n, p); dim(x)
beta = rnorm(nzc)
fx = x[, seq(nzc)] %*% beta
eps = rnorm(n) * 5
y = drop(fx + eps)
cvfit = cv.glmnet(x, y)
coef(cvfit, s = cvfit$lambda.min)
```

nzc=10

- In the simulated data, the first 10 variables are relevant variables.
- The selected variables include all the relevant variables but variable 3. In addition to the relevant variables, the selected variables also include 10 irrelevant variables.

69. Final remarks on model selectoin

- The true mechanism behind the data is too complicated to model.
- Any model is only an approximation to the truth, good or bad.

  All models are wrong, but some are useful. – *George Box.*

- Different model selection procedures do not necessarily select the same model.
- Different criteria do not necessarily select the same model.
- If different models are selected by different procedures or different criteria, each of them should be entertained.